Ethics in Software Engineering

Lecture 13

Agenda

- What is ethics?
- The Software Engineering Code of Ethics and Professional Practice
- A Guide to Use the Code (but with ACM Code of Ethics and Professional Conduct)
- Exemplars Case Studies
- Some Real Cases: Volskwagen and Boeing

What is ethics?

What is ethics?

- Ethics is the study of what is morally right and wrong, or a set of beliefs about what is morally right and wrong.
  - Cambridge Dictionary
- Ethics is concerned about norms for conduct that distinguish between acceptable and unacceptable behaviour.
- Today, the study of ethics can be found in many different places. As an academic field of study, it belongs primarily to the discipline of philosophy.

What is ethics?

- On a personal level, it can be expressed in an individual's self-reflection and continual strivings to become a better person.
- In work life, it is often formulated in formal codes or standards to which all members of a profession are held, such as those of medical ethics.
- As professionals, software engineers should abide by a code of ethics that guides the work that they do and the products that they produce.

Unethical or Illegal?
Some Real Cases: Volskwagen and Boeing

Volkswagen se enfrenta a la peor crisis de su historia reciente. El mayor fabricante de automóviles del mundo ha reconocido que instaló un *software* para esquivar controles medioambientales en 11 millones de vehículos diésel de todo el mundo. Estas son algunas de las claves del escándalo que ha socavado la imagen del gigante automovilístico alemán:

La agencia medioambiental estadounidense (EPA) acusa a Volkswagen el pasado viernes de haber hecho trampa al instalar en 482.000 vehículos de Estados Unidos de manera deliberada un programa informático diseñado para evitar los límites a las emisiones. El programa detecta el momento en que el vehículo está siendo sometido a una prueba y disminuye la emisión de gases contaminantes solo en ese momento. Una vez en circulación, los vehículos llegaban a emitir óxidos de nitrógeno (NOx) hasta 40 veces superiores a lo permitido en Estados Unidos.

El impacto económico para la empresa se cifra inicialmente en un máximo de 18.000 millones de dólares (16.000 millones de euros), resultado de una multa máxima de 37.500 dólares por vehículo, si bien los analistas estiman poco probable que se aplique una sanción de dichas proporciones. Las acciones del grupo se desploman un 18,6%.

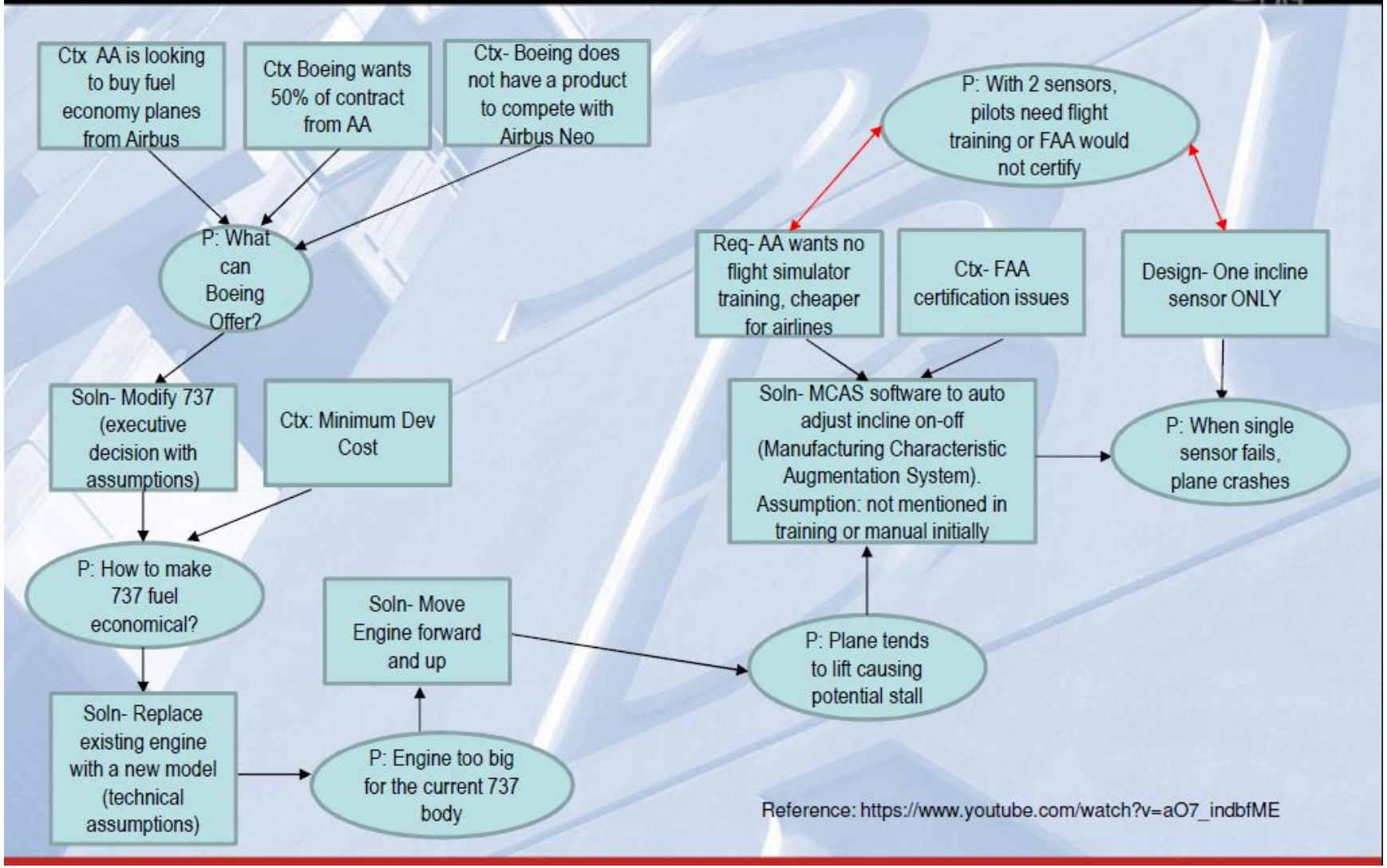
Volkswagen reconoce el error y detiene la comercialización. La compañía admite que implantó el *software* tras la acusación de la EPA. "Lamento profundamente haber decepcionado a nuestros clientes y a la opinión pública", señala el ahora cuestionado presidente de Volkswagen, Martin Winterkorn. El responsable de Volkswagen en Estados Unidos es más directo: "La hemos cagado por completo", dice tras admitir el engaño. La

MÁS INFORMACIÓN

- Volkswagen reconoce que trucó 11 millones de coches en todo el mundo
- "La hemos cagado por completo", dice el jefe de Volkswagen en EE UU
- Alemania investigará a Volkswagen por falsear sus emisiones en EE UU
- Volkswagen admite haber engañado a EE UU con la emisión de gases

Fuente: https://elpais.com/economia/2015/09/22/actualidad/1442929060_472526.html

737-Max Case – based on 60 Minutes Report



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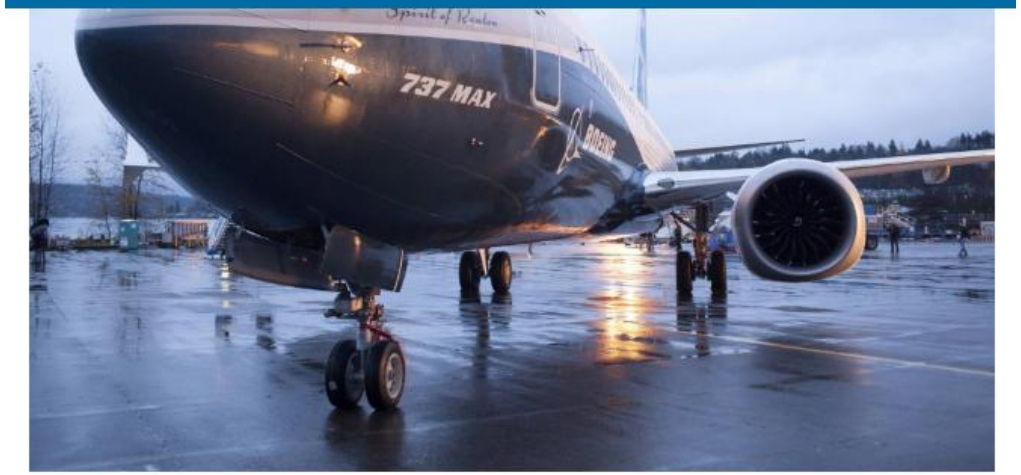


Imagen de un Boeing 737 MAX 8, en la planta de Boeing en Renton, Washington , en 2015. MATT MCKNIGHT (REUTERS)

Según un informe oficial del pasado 4 de abril, no se ha detectado ningún fallo humano entre las causas del accidente de aviación que, el pasado 10 de marzo, se cobró 157 vidas, al caer en picado a los pocos minutos de despegar. El informe detalla que los pilotos trataron de desactivar el software conocido como MCAS (siglas en inglés de "Sistema de Aumento de las Características de Maniobra") hasta en cuatro ocasiones, sin conseguirlo. El modelo 737 MAX fue diseñado para mejorar la potencia y eficiencia del Boeing 737. Los nuevos motores son más pesados que sus predecesores, y fueron colocados en distinta posición, lo que modificó el centro de gravedad. El software MCAS se diseñó para que actuara automáticamente cuando un sensor de "ángulo de ataque" detectase que la inclinación del morro es muy alta, forzando a inclinarlo hacia abajo. Todo indica que un sensor envió información errónea al software MCAS. Aunque el fallo de un sensor puede considerarse un fallo técnico, el software que actúa en base a la información que le transmiten dispositivos externos, como sensores, debe tener mecanismos de recuperación de fallos de dichos dispositivos o, como mínimo, permitir su propia desactivación. Por tanto, todo indica que lo que se produjo fue un fatal error de software que impidió la desactivación del MCAS.

El accidente aéreo del vuelo ET302 de Ethiopian Airlines, en el que han muerto 157 personas, ha desencadenado una crisis sin precedentes en el fabricante Boeing, quien ha visto **cómo uno de sus modelos estrella, el 737 Max, ha sufrido dos siniestros fatales** en menos de cinco meses.

Las similitudes entre el accidente de Etiopía y el de Lion Air que se produjo en Indonesia en octubre de 2018 son demasiadas, lo que de inmediato ha puesto bajo sospecha a la familia de aviones 737 Max, que fue lanzada al mercado en 2017 y **representa la última variante de la línea más vendida de la historia**, desde que el primer 737 viese la luz en 1967.

La lista de países y de compañías que han suspendido temporalmente los vuelos operados por el 737 Max no ha dejado de crecer en las últimas horas. A ella se ha sumado también la Unión Europea, que ha cerrado su espacio aéreo a este modelo.

Fuente: http://www.rtve.es/noticias/20190312/737-max-exito-convertido-pesadilla-hace-temblar-boeing/1900621.shtml

actualidad.rt.com/actualidad/453002-reporte-investigadores-culpa-boeing-accidente

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Determinan la culpabilidad de Boeing en el accidente aéreo de 2019 en Etiopía

Publicado: 24 dic 2022 16:05 GMT

Un sensor falló poco después del despegue y sus datos erróneos afectaron a un sistema desarrollado por el fabricante, hasta que los pilotos perdieron el control del avión, concluyeron los investigadores.



actualidad.rt.com/actualidad/379481-boeing-multa-pagos-737max

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Boeing pagará 2.500 millones de dólares para evitar cargos de conspiración sobre los 737 MAX

Publicado: 8 ene 2021 03:46 GMT

La compañía ha llegado a un acuerdo con el Departamento de Justicia de EE.UU. para resolver un cargo penal en su contra.



Boeing pagará 2.500 millones de dólares para zanjar la investigación del Departamento de Justicia de Estados Unidos y cargos por conspiración para engañar a los reguladores federales sobre cuestiones de seguridad de sus aviones 737 MAX, modelos que protagonizaron dos catástrofes aéreas poco después de entrar en servicio.

Parte de esta suma, 243,6 millones de dólares, es la multa, mientras que el resto son los pagos a los familiares de las víctimas de los accidentes y a las aerolíneas que adquirieron los 737 MAX.

Según los fiscales, los empleados de Boeing emitieron declaraciones engañosas y dijeron medias verdades sobre asuntos de seguridad de los aviones de este modelo a los representantes de la Administración Federal de Aviación, y luego encubrieron sus acciones.



"Defectuoso e inseguro": Un informe del Congreso de EE.UU. describe fallos que llevaron a los siniestros del Boeing 737 MAX

Si Boeing cumple con los términos del acuerdo, tres años después el Gobierno retirará el cargo penal de conspiración para defraudar a las autoridades contra el icónico fabricante de aviones.

"Este histórico acuerdo de enjuiciamiento diferido servirá para siempre como un duro recordatorio de la importancia primordial de la seguridad en la industria de la aviación comercial, y de que la integridad y la transparencia nunca pueden ser sacrificadas por la eficiencia o el beneficio", declaró la agente especial a cargo Andrea M. Kropf, del Departamento de Transporte de la Oficina del Inspector General para la Región Medio Oeste.

Los vuelos de los Boeing 737 MAX fueron suspendidos en todo el mundo en marzo de 2019, después de dos siniestros aéreos. Aquel mes, 157 personas murieron en un accidente de Ethiopian Airlines. En octubre de 2018, se estrelló otro 737 MAX, de la aerolínea indonesia Lion Air, cobrándose la vida de 189 personas. Ambas catástrofes aéreas se produjeron debido al funcionamiento incorrecto del sistema MCAS.

The Software Engineering Code of Ethics and Professional Practice

The Software Engineering Code of Ethics and Professional Practice

- An ACM/IEEE-CS Joint Task Force has produced a Software Engineering Code of Ethics and Professional Practices (Version 5.2). The code states:
  - Software engineers shall commit themselves to making the analysis, specification, design, development, testing and maintenance of software a beneficial and respected profession.
- In accordance with their commitment to the health, safety and welfare of the public, software engineers shall adhere to the following Eight Principles:

Principles of the Software Engineering Code of Ethics

1. PUBLIC – Software engineers shall act consistently with the public interest.
2. CLIENT AND EMPLOYER – Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.
3. PRODUCT – Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
4. JUDGMENT – Software engineers shall maintain integrity and independence in their professional judgment.

Principles of the Software Engineering Code of Ethics

5. MANAGEMENT – Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6. PROFESSION – Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7. COLLEAGUES – Software engineers shall be fair to and supportive of their colleagues.
8. SELF – Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Principle 1: Public

Software engineers shall act consistently with the public interest. In particular, software engineers shall, as appropriate:

1.01. Accept full responsibility for their own work.

1.02. Moderate the interests of the software engineer, the employer, the client and the users with the public good.

1.03. Approve software only if they have a well-founded belief that it is safe, meets specifications, passes appropriate tests, and does not diminish quality of life, diminish privacy or harm the environment. The ultimate effect of the work should be to the public good.

1.04. Disclose to appropriate persons or authorities any actual or potential danger to the user, the public, or the environment, that they reasonably believe to be associated with software or related documents.

1.05. Cooperate in efforts to address matters of grave public concern caused by software, its installation, maintenance, support or documentation.

1.06. Be fair and avoid deception in all statements, particularly public ones, concerning software or related documents, methods and tools.

1.07. Consider issues of physical disabilities, allocation of resources, economic disadvantage and other factors that can diminish access to the benefits of software.

1.08. Be encouraged to volunteer professional skills to good causes and contribute to public education concerning the discipline.

Principle 2: Client and Employer

Software engineers shall act in a manner that is in the best interests of their client and employer, consistent with the public interest. In particular, software engineers shall, as appropriate:

2.01. Provide service in their areas of competence, being honest and forthright about any limitations of their experience and education.

2.02. Not knowingly use software that is obtained or retained either illegally or unethically.

2.03. Use the property of a client or employer only in ways properly authorized, and with the client's or employer's knowledge and consent.

2.04. Ensure that any document upon which they rely has been approved, when required, by someone authorized to approve it.

2.05. Keep private any confidential information gained in their professional work, where such confidentiality is consistent with the public interest and consistent with the law.

2.06. Identify, document, collect evidence and report to the client or the employer promptly if, in their opinion, a project is likely to fail, to prove too expensive, to violate intellectual property law, or otherwise to be problematic.

2.07. Identify, document, and report significant issues of social concern, of which they are aware, in software or related documents, to the employer or the client.

2.08. Accept no outside work detrimental to the work they perform for their primary employer.

2.09. Promote no interest adverse to their employer or client, unless a higher ethical concern is being compromised; in that case, inform the employer or another appropriate authority of the ethical concern.

Principle 3: Product

Software engineers shall ensure that their products and related modifications meet the highest professional standards possible. In particular, software engineers shall, as appropriate:

3.01. Strive for high quality, acceptable cost and a reasonable schedule, ensuring significant tradeoffs are clear to and accepted by the employer and the client, and are available for consideration by the user and the public.

3.02. Ensure proper and achievable goals and objectives for any project on which they work or propose.

3.03. Identify, define and address ethical, economic, cultural, legal and environmental issues related to work projects.

3.04. Ensure that they are qualified for any project on which they work or propose to work by an appropriate combination of education and training, and experience.

3.05. Ensure an appropriate method is used for any project on which they work or propose to work.

3.06. Work to follow professional standards, when available, that are most appropriate for the task at hand, departing from these only when ethically or technically justified.

3.07. Strive to fully understand the specifications for software on which they work.

3.08. Ensure that specifications for software on which they work have been well documented, satisfy the users' requirements and have the appropriate approvals.

3.09. Ensure realistic quantitative estimates of cost, scheduling, personnel, quality and outcomes on any project on which they work or propose to work and provide an uncertainty assessment of these estimates.

3.10. Ensure adequate testing, debugging, and review of software and related documents on which they work.

3.11. Ensure adequate documentation, including significant problems discovered and solutions adopted, for any project on which they work.

3.12. Work to develop software and related documents that respect the privacy of those who will be affected by that software.

3.13. Be careful to use only accurate data derived by ethical and lawful means, and use it only in ways properly authorized.

3.14. Maintain the integrity of data, being sensitive to outdated or flawed occurrences.

3.15 Treat all forms of software maintenance with the same professionalism as new development.

Principle 4: Judgement

Software engineers shall maintain integrity and independence in their professional judgment. In particular, software engineers shall, as appropriate:

4.01. Temper all technical judgments by the need to support and maintain human values.

4.02 Only endorse documents either prepared under their supervision or within their areas of competence and with which they are in agreement.

4.03. Maintain professional objectivity with respect to any software or related documents they are asked to evaluate.

4.04. Not engage in deceptive financial practices such as bribery, double billing, or other improper financial practices.

4.05. Disclose to all concerned parties those conflicts of interest that cannot reasonably be avoided or escaped.

4.06. Refuse to participate, as members or advisors, in a private, governmental or professional body concerned with software related issues, in which they, their employers or their clients have undisclosed potential conflicts of interest.

Principle 5: Management

Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance . In particular, those managing or leading software engineers shall, as appropriate:

5.01 Ensure good management for any project on which they work, including effective procedures for promotion of quality and reduction of risk.

5.02. Ensure that software engineers are informed of standards before being held to them.

5.03. Ensure that software engineers know the employer's policies and procedures for protecting passwords, files and information that is confidential to the employer or confidential to others.

5.04. Assign work only after taking into account appropriate contributions of education and experience tempered with a desire to further that education and experience.

5.05. Ensure realistic quantitative estimates of cost, scheduling, personnel, quality and outcomes on any project on which they work or propose to work, and provide an uncertainty assessment of these estimates.

5.06. Attract potential software engineers only by full and accurate description of the conditions of employment.

5.07. Offer fair and just remuneration.

5.08. Not unjustly prevent someone from taking a position for which that person is suitably qualified.

5.09. Ensure that there is a fair agreement concerning ownership of any software, processes, research, writing, or other intellectual property to which a software engineer has contributed.

5.10. Provide for due process in hearing charges of violation of an employer's policy or of this Code.

5.11. Not ask a software engineer to do anything inconsistent with this Code.

5.12. Not punish anyone for expressing ethical concerns about a project.

Principle 6: Profession

Software engineers shall advance the integrity and reputation of the profession consistent with the public interest. In particular, software engineers shall, as appropriate:

6.01. Help develop an organizational environment favorable to acting ethically.

6.02. Promote public knowledge of software engineering.

6.03. Extend software engineering knowledge by appropriate participation in professional organizations, meetings and publications.

6.04. Support, as members of a profession, other software engineers striving to follow this Code.

6.05. Not promote their own interest at the expense of the profession, client or employer.

6.06. Obey all laws governing their work, unless, in exceptional circumstances, such compliance is inconsistent with the public interest.

6.07. Be accurate in stating the characteristics of software on which they work, avoiding not only false claims but also claims that might reasonably be supposed to be speculative, vacuous, deceptive, misleading, or doubtful.

6.08. Take responsibility for detecting, correcting, and reporting errors in software and associated documents on which they work.

6.09. Ensure that clients, employers, and supervisors know of the software engineer's commitment to this Code of ethics, and the subsequent ramifications of such commitment.

6.10. Avoid associations with businesses and organizations which are in conflict with this code.

6.11. Recognize that violations of this Code are inconsistent with being a professional software engineer.

6.12. Express concerns to the people involved when significant violations of this Code are detected unless this is impossible, counter-productive, or dangerous.

6.13. Report significant violations of this Code to appropriate authorities when it is clear that consultation with people involved in these significant violations is impossible, counter-productive or dangerous.

Principle 7: Colleagues

Software engineers shall be fair to and supportive of their colleagues. In particular, software engineers shall, as appropriate:

7.01. Encourage colleagues to adhere to this Code.

7.02. Assist colleagues in professional development.

7.03. Credit fully the work of others and refrain from taking undue credit.

7.04. Review the work of others in an objective, candid, and properly-documented way.

7.05. Give a fair hearing to the opinions, concerns, or complaints of a colleague.

7.06. Assist colleagues in being fully aware of current standard work practices including policies and procedures for protecting passwords, files and other confidential information, and security measures in general.

7.07. Not unfairly intervene in the career of any colleague; however, concern for the employer, the client or public interest may compel software engineers, in good faith, to question the competence of a colleague.

7.08. In situations outside of their own areas of competence, call upon the opinions of other professionals who have competence in that area.

Principle 8: Self

Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession. In particular, software engineers shall continually endeavor to:

8.01. Further their knowledge of developments in the analysis, specification, design, development, maintenance and testing of software and related documents, together with the management of the development process.

8.02. Improve their ability to create safe, reliable, and useful quality software at reasonable cost and within a reasonable time.

8.03. Improve their ability to produce accurate, informative, and well-written documentation.

8.04. Improve their understanding of the software and related documents on which they work and of the environment in which they will be used.

8.05. Improve their knowledge of relevant standards and the law governing the software and related documents on which they work.

8.06 Improve their knowledge of this Code, its interpretation, and its application to their work.

8.07 Not give unfair treatment to anyone because of any irrelevant prejudices.

8.08. Not influence others to undertake any action that involves a breach of this Code.

8.09. Recognize that personal violations of this Code are inconsistent with being a professional software engineer.

Concluding Comments

- Although each of these eight principles is equally important, an overriding theme appears: a software engineer should work in the public interest.
- On a personal level, a software engineer should abide by the following rules:
  - Never steal data for personal gain.
  - Never distribute or sell proprietary information obtained as part of your work on a software project.
  - Never maliciously destroy or modify another person's programs, files, or data.
  - Never violate the privacy of an individual, a group, or an organization.
  - Never hack into a system for sport or profit.
  - Never create or promulgate a computer virus or worm.
  - Never use computing technology to facilitate discrimination or harassment.

Broadening Our View of Software Engineering Ethics

- According to Shannon Vallor, software engineers need to broaden their understanding of a number of aspects related to their ethics concerns such as:
  - The types of harms the public can suffer as result of their work;
  - How software engineers contribute to the good life for others;
  - Who exactly are the 'public' to whom the engineer is obligated;
  - Why the software engineer is obligated to protect the public;
  - What other ethical obligations software engineers are under;
  - How software engineers can actually live up to ethical standards;
  - What is the end goal of an ethical life in software engineering;
  - What are the professional codes of software engineering ethics;

A Guide to Use the Code (but with ACM Code of Ethics and Professional Conduct)

A Guide to Use the Code

www.acm.org/code-of-ethics/case-studies

Home > Code Of Ethics > Code Of Ethics Case Studies

ACM Code of Ethics and Professional Conduct

Using the Code

Case Studies

The ACM Code of Ethics and Professional Practice ("the Code") is meant to inform practice and education. It is useful as the conscience of the profession, but also for individual decision-making.

As prescribed by the Preamble of the Code, computing professionals should approach the dilemma with a holistic reading of the principles and evaluate the situation with thoughtful consideration to the circumstances. In all cases, the computing professional should defer to the public good as the paramount consideration. The analyses in the following cases highlight the intended interpretations of members of the 2018 Code task force, and should help guide computing professionals in how to apply the Code to various situations.



Case Study: Malware

Rogue Services touts its web hosting as "cheap, guaranteed uptime, no matter what." While some of Rogue's clients are independent web-based retailers, most are focused on malware and spam, which leverage Rogue for continuous delivery. Corrupted advertisements often link to code hosted on Rogue to exploit browser vulnerabilities to infect machines with ransomware. Rogue refuses to intervene with



Case Study: Medical Implants

Corazón is a medical technology startup that builds implantable heart health monitoring devices. After being approved by multiple countries' medical device regulation agencies, Corazón quickly gained market share based on the ease of use of the app and the company's vocal commitment to securing patients' information. Corazón also worked with several charities to provide free or reduced access to patients living below the



Guiding Members with a Framework of Ethical Conduct

Learn more about ACM's commitment to ethical standards: the ACM Code of Ethics, Software Engineering Code of Ethics and Professional Practice, and Committee on Professional Ethics (COPE), which is guiding these and other intiatives.



Ask an Ethicist

Ask an Ethicist invites ethics questions related

Using the Code: Medical Implant Risk Analysis

Corazón is a medical technology startup that builds an implantable heart health monitoring device. The device comes with a smart phone app that can monitor and control the device wirelessly, as well as storing a persistent record that can be shared with medical providers. After being approved by multiple countries' medical device regulation agencies, Corazón quickly gained market share based on the ease of use of the app and the company's vocal commitment to securing patients' information. To further expand their impact, Corazón worked with several charities to provide free or reduced access to patients living below the poverty line.

As a basic security mechanism, Corazón's implant could only be accessible through short-range wireless connections, requiring the phone and implant to be in close proximity. Data transferred between the app and the device employed standard cryptographic algorithms, and all data stored locally on the phone was encrypted. To support on-going improvement, Corazón had an open bug bounty program inviting disclosure of potential vulnerabilities in their app.

At a recent security conference, an independent researcher claims to have found a vulnerability in the wireless connectivity. The researcher presents a proof-of-concept demonstration where a second device in close proximity could modify commands sent to the implant to force a device reset. The attack relied on the use of a hard-coded initialization value stored in the implant device that created a predictable pattern in the data exchanges that could be manipulated. In consultation with Corazón's technical leaders, the researcher concludes that the risk of harm with this attack is negligible, given the limited capabilities of the device.

Analysis

Corazón's practices embody the goals of several principles in the Code. Corazón's products and their charity work contribute to society and to human well-being, consistent with the aims of Principle 1.1. By working within the regulations of governmental agencies, Corazón demonstrated a commitment to Principle 2.3. Corazón's use of cryptography and vulnerability disclosure adheres to the robust security goals of Principle 2.9. Furthermore, Corazón's reliance on standard cryptographic algorithms—rather than attempting to devise an unproven proprietary technique—shows commitment to Principle 2.6, restricting their developers' work to areas of competence.

Corazón's consultation with the researcher also highlights a key aspect of Principle 2.5. The design and implementation of Corazón's products exhibit a commitment to comprehensive and thorough risk analysis. Furthermore, Corazón welcomed independent security evaluation to identify additional issues that their designers overlooked. Once a potential vulnerability was discovered, Corazón acted responsibly and quickly to determine the scope of the flaw with the aim of mitigating the harm.

One area of concern regarding Corazón's design is the use of a hard-coded value in the implant. Given the nature of the device, fixing this design choice would be difficult if it proved necessary. However, there is insufficient evidence at this point to determine the scope of the risk induced by this design.

Corazón's on-going commitment to security and improvement also exemplify an important aspect of Principle 3.7. Corazón's rapid success in this specialized healthcare field is an instance of the integration of technology into the infrastructure of society. Recognizing the increased stewardship required by this Principle, Corazón began working with charities to serve individuals whose poverty may have excluded them from access.

Exemplars Case Studies

Case Study 1

- George Babbage is an experienced software developer working for Acme Software Company. Mr. Babbage is now working on a project for the U.S. Department of Defence, testing the software used in controlling an experimental jet fighter. George is the quality control manager for the software. Early simulation testing revealed that, under certain conditions, instabilities would arise that could cause the plane to crash. The software was patched to eliminate the specific problems uncovered by the tests. After these repairs, the software passed all the simulation tests.
- George is not convinced that the software is safe. He is worried that the problems uncovered by the simulation testing were symptomatic of a design flaw that could only be eliminated by an extensive redesign of the software. He is convinced that the patch that was applied to remedy the specific tests in the simulation did not address the underlying problem. But, when George brings his concerns to his superiors, they assure him that the problem has been resolved. They further inform George that any major redesign effort would introduce unacceptable delays, resulting in costly penalties to the company.
- There is a great deal of pressure on George to sign off on the system and to allow it to be flight tested. It has even been hinted that, if he persists in delaying the system, he will be fired. What should George do next?

Case Study 1: Relevant Clauses

- Principle 1. **PUBLIC** Software engineers shall act consistently with the public interest. In particular, software engineers shall, as appropriate:
  - 1.03. Approve software only if they have a well-founded belief that it is safe, meets specifications, passes appropriate tests, and does not diminish quality of life, diminish privacy or harm the environment. The ultimate effect of the work should be to the public good.
  - 1.04. Disclose to appropriate persons or authorities any actual or potential danger to the user, the public, or the environment, that they reasonably believe to be associated with software or related documents.

Case Study 1: Relevant Clauses

- Principle 3. **PRODUCT** Software engineers shall ensure that their products and related modifications meet the highest professional standards possible. In particular, software engineers shall, as appropriate:
  - 3.10. Ensure adequate testing, debugging, and review of software and related documents on which they work.

Case Study 1: Relevant Clauses

- Principle 5. **MANAGEMENT** Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance. In particular, those managing or leading software engineers shall, as appropriate:
  - 5.01 Ensure good management for any project on which they work, including effective procedures for promotion of quality and reduction of risk.
  - 5.11. Not ask a software engineer to do anything inconsistent with this Code.

Case Study 2

- Mauricio copia varios archivos de un computador del laboratorio. El revisa todos los archivos y encuentra un proyecto de software muy interesante, al cual decide hacer ciertos arreglos.
- A Mauricio le proponen participar en un concurso de desarrollo de software y él acepta. El decide presentar el proyecto mejorado como suyo. Sin embargo, el autor del proyecto, David también está participando en el concurso. Por lo tanto, existen 2 proyectos similares y ambos autores son llamados para justificar dicha situación. Ambos argumentan que los proyectos son suyos ¿quién tiene la razón? Explique.

Case Study 3

- Federico trabajaba en una empresa desarrolladora de software y todo lo que él desarrolló pertenecía a la empresa, así lo estipula el reglamento de dicha empresa con la cual firmó un contrato con esas condiciones.
- Federico trabaja actualmente en otra empresa -mucho más grande- en el área de servicio al cliente.
- Él se da cuenta que puede facilitar su trabajo si utiliza una de las aplicaciones que desarrolló en la antigua empresa, pero no está seguro de utilizarla. Comenta sobre su dilema a un amigo y colega Mario, quien le sugiere que no lo utilice porque no es ético. Federico aun así decide utilizar la aplicación.
- Luego de un tiempo, el jefe de Federico nota que la aplicación sirve para otras áreas y le pide que la adapte, de lo contrario será despedido. ¿Qué debe hacer Federico?

Next On

Take Away Points

- What ethics is.
- The eight Principles contained in the Code related to the behaviour of and decisions made by professional software engineers.
- The Code is not a simple ethical algorithm that generates ethical decisions. In some situations, standards may be in tension with each other or with standards from other sources. These situations require the software engineer to use ethical judgment to act in a manner which is most consistent with the spirit of the Code of Ethics and Professional Practice, given the circumstances.

Further Reading

- Pressman and Maxim, “Software Engineering”
  - Chapter 39: Concluding Comments
- Ian Somerville, “Software Engineering”
  - Chapter 1: Introduction
- Shannon Vallor, “An Introduction to Software Engineering Ethics”
- Unger, S.H., 2000. Examples of real world engineering ethics problems. Science and engineering ethics, 6(3), pp.423-430.
- ACM, Online. https://ethics.acm.org/code-of-ethics/software-engineering-code
- ACM, Online. ACM Code of Ethics and Professional Conduct: Using the Code. https://www.acm.org/code-of-ethics/case-studies

Next Lecture

- Concepts of Software Architecture and Architectural Styles